Holiday Flight-Service Patterns at Shanghai Hongqiao Airport

A calendar-aware descriptive analysis of China's 2026 National Day period

Transportation Engineering Research Paper

Study period: September 22-October 22, 2026

Focus period: October 1-7, 2026

Author: [Add your name]

Course: [Add course name and code]

Submission date: [Add date]

Scope note: This is an exploratory single-airport study of recorded flight events. The file does not contain passenger counts, traveler motives, or measured emissions.

Abstract

This study characterizes flight-event patterns at Shanghai Hongqiao International Airport (SHA) during China's National Day holiday, October 1-7, 2026, using 23,705 records across 31 local dates. It addresses a calendar-classification problem in short-window transport analysis: the adjacent seven-day periods share weekdays with the holiday week but contain different official day types, including Mid-Autumn public holidays before the study period and a Saturday adjusted workday after it. The National Day week contains 5,381 records (768.7 per day), compared with 764.4 and 732.3 records per day in the preceding and following windows. The single adjusted workday, October 10, contains 682 records; 4.41% of its known-status records are marked canceled. The holiday-week cancellation share is 1.91%, compared with 3.74% before and 2.41% after. The analysis contributes a calendar-aware descriptive baseline for airport service patterns, an explicit account of how make-up workdays affect comparison-window interpretation, and a data-boundary framework separating flight-event supply from passenger demand, validated performance, and emissions. Because the dataset contains no passenger, alternative-mode, or verified emissions measures, results are descriptive and do not establish traveler behavior or sustainability impacts.

Keywords: air transportation; holiday travel; adjusted workday; flight-service patterns; calendar effects; Shanghai Hongqiao

1. Introduction and problem statement

Holiday periods are planning-relevant because travel activity and transport-service provision may vary with the timing of public holidays, work schedules, and the surrounding calendar. In China, an extended public holiday can be paired with officially designated make-up workdays on weekends. The calendar day of week is therefore not, on its own, a sufficient descriptor of whether a date is a working day or a holiday. Treating a make-up workday as an ordinary Saturday can confound comparisons of flight-service activity and lead to misleading conclusions about holiday-related change.

A second problem is measurement. Airport flight-event records describe scheduled services and status labels, but they are not passenger counts and do not directly measure demand, airport capacity, congestion, or environmental impact. Transport planning and sustainability assessments require outcomes and system boundaries matched to the question (IPCC, 2022; ITF, 2023). If event records are interpreted as passenger behavior, or schedule counts as realized throughput, the analysis exceeds what the data can support.

The study addresses these problems through a calendar-aware descriptive analysis of Shanghai Hongqiao Airport (SHA). It compares the October 1-7, 2026 National Day period with weekday-aligned adjacent windows, examines scheduled-hour and directional-route profiles, and reports status-based cancellation proportions. Calendar classifications from the supplied dataset are interpreted against the official 2026 holiday schedule (General Office of the State Council, 2025). The term 'flight behavior' is operationalized as flight-service patterns; the analysis does not infer individual traveler choices.

2. Objectives, research questions, and contribution

The objective is to describe how recorded airport service activity is distributed around the National Day holiday and to establish whether the holiday calendar, including an adjusted workday, changes the interpretation of adjacent-period comparisons.

RQ1. How does the number of recorded arrival and departure flight events per day during October 1-7 compare with the adjacent seven-day periods?

RQ2. Do the distribution of scheduled flight times and the most frequently recorded directional routes differ during the holiday week?

RQ3. How do recorded cancellation proportions differ across the pre-holiday, holiday, and post-holiday windows?

RQ4. What flight-event pattern is recorded on the adjusted workday immediately after the National Day holiday, and how does it compare descriptively with other calendar-day types in the sample?

The questions are descriptive, not causal. The study makes three contributions. First, it provides a reproducible, single-airport baseline of daily event volumes, scheduled-hour shares, route frequencies, and status-based cancellations for a Chinese National Day period. Second, it explicitly incorporates official calendar day type, showing why weekday-aligned windows are not necessarily comparable when they include public holidays or make-up workdays. Third, it sets clear interpretive boundaries between service-event records and unobserved passenger demand, validated operational performance, and sustainability outcomes. The contribution is a transparent case analysis and measurement framework, not a causal estimate or a general model of Chinese holiday travel.

3. Study context, data, and methods

The analysis uses the supplied sha_master.csv file. It contains 23,705 unique event_id records and covers 31 local dates, from 2026-09-22 through 2026-10-22. Each record describes an arrival or departure event at SHA, with fields for local date and hour, directional route, displayed status, and calendar classification. The seven-day study windows are September 24-30 (pre-National Day), October 1-7 (National Day), and October 8-14 (post-National Day). They are equal in length and begin on the same weekday, but their official calendar types differ: September 25-27 are Mid-Autumn public holidays, October 1-7 are National Day public holidays, and October 10 is designated as an adjusted workday in the official 2026 holiday schedule (General Office of the State Council, 2025). The schedule also designates September 20 as a workday, but that date falls outside the supplied dataset. Thus weekday alignment does not make these periods fully calendar-equivalent.

The primary volume measure is the count of event records per local date, split by arrival and departure. For the comparison table, the daily mean is total event records divided by the seven dates. A canceled event remains in the event-record volume because it was part of the listed service, but it is not an operated flight. Route frequencies count directional event records. The hourly plot compares each window's share of all its records by scheduled local hour, avoiding direct comparisons of unequal totals. The calendar-type plot groups daily totals by the dataset's calendar_day_type field and shows each date separately. The adjusted-workday category contains one date only, so it is presented as a single-case observation rather than a general estimate.

Cancellation proportion is the number of records whose status is 'Canceled' divided by records with a nonblank status other than 'Unknown'. Rows with blank or unknown status are excluded from this denominator and reported as missing status. No inferential statistical test is used: there are only three short, adjacent windows and no control for weather, capacity, schedule changes, or other concurrent conditions. The analysis is reproducible from the included Python script.

Ethics and privacy: the report aggregates operational event data and does not reproduce flight identifiers or attempt to infer personal travel histories. Any further work using passenger-level or booking data would require appropriate access controls and ethical review.

4. Results

Across the seven holiday dates, the file contains 5,381 flight-event records: 2,699 arrivals and 2,682 departures. The mean is 768.7 records per day, a 0.6% difference from the pre-National Day daily mean and a 5.0% difference from the post-National Day mean. These values are similar to the preceding week and higher than the following week in this sample; they should not be treated as passenger demand or airport throughput.

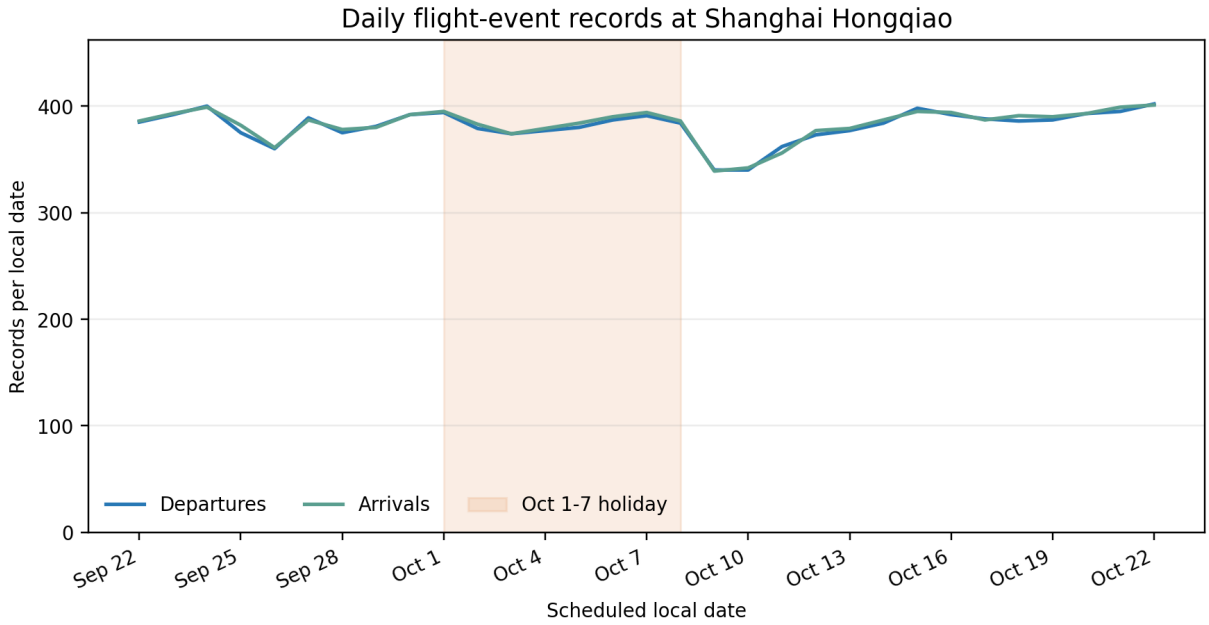


Figure 1. Daily arrival and departure event records in the full dataset. Shading marks October 1-7. Records include events displayed as canceled and therefore are not counts of completed flights.

Figure 1 places the holiday week within the full 31-date observation period. Daily counts vary, while arrivals and departures remain close in number. The display supports inspection of short-run variation but does not identify whether it reflects holiday travel, airline scheduling, data capture, or operational disruptions.

Seven-day window	Flight-event records	Records / day	Arrivals	Departures	Canceled / known
Pre-holiday	5,351	764.4	2,679	2,672	200 / 5,348 (3.74%)
National Day	5,381	768.7	2,699	2,682	103 / 5,380 (1.91%)
Post-holiday	5,126	732.3	2,566	2,560	123 / 5,097 (2.41%)

Note. Status rates use known-status records as the denominator; blank and 'Unknown' statuses are excluded. Flight-event records include canceled records and are not passenger totals.

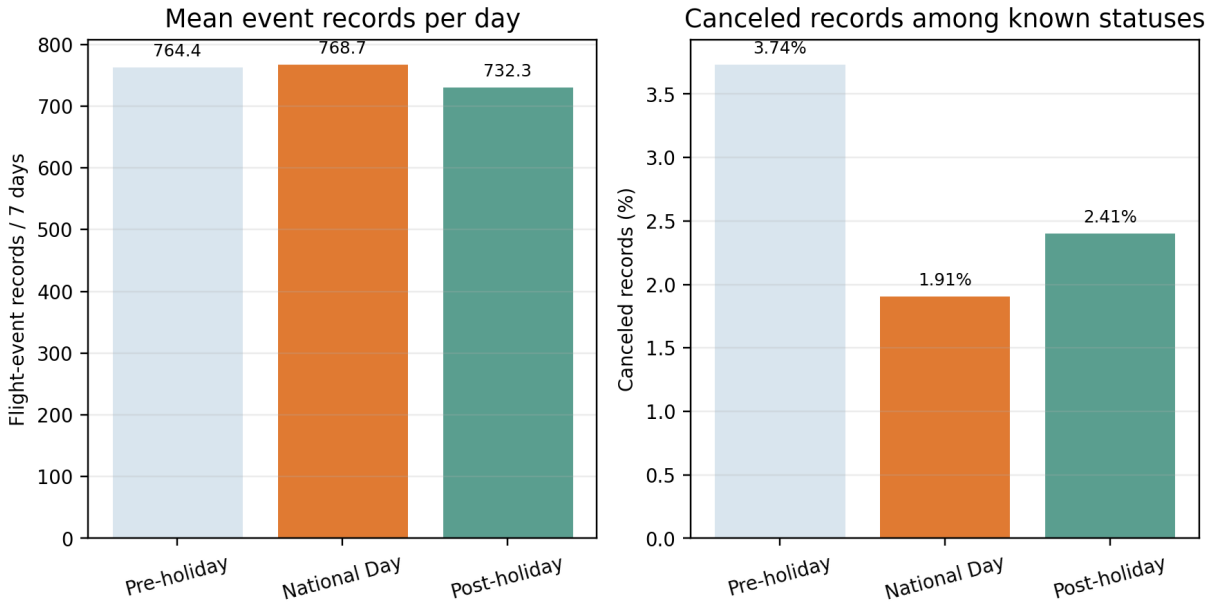


Figure 2. Equal-length, weekday-aligned windows. The right panel uses known-status records only; the count of known records differs by period. Calendar types also differ across windows.

The holiday period has 103 canceled records among 5,380 known-status records (1.91%). The corresponding proportions are 200 of 5,348 (3.74%) before the holiday and 123 of 5,097 (2.41%) afterward. The direction of this descriptive difference should be interpreted cautiously: 29 post-holiday records have blank or unknown status, compared with 3 before and 1 during the holiday, and the data do not explain the reasons for cancellation.

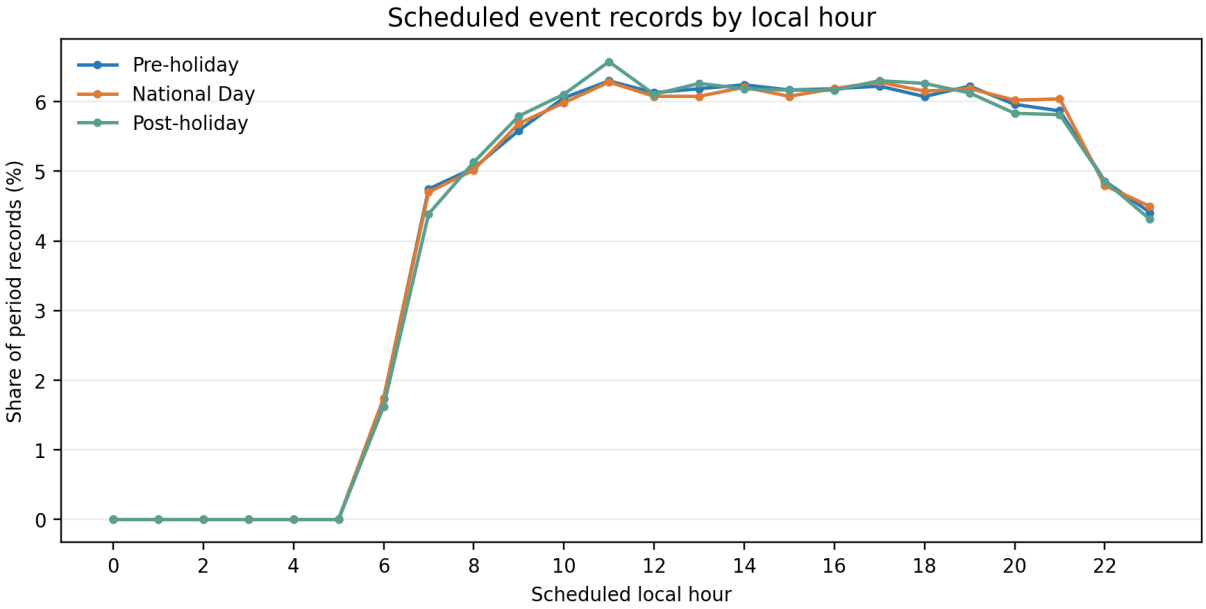


Figure 3. Each line shows the percentage distribution of event records by scheduled local hour within a seven-day window. The profile describes schedule timing, not passenger arrival times or congestion.

The hourly profiles make it possible to assess whether records are concentrated in different parts of the day, while normalizing for each window's total. This is a schedule-composition measure. No conclusion about queues or congestion follows without airport capacity, runway, taxi, and actual movement data.

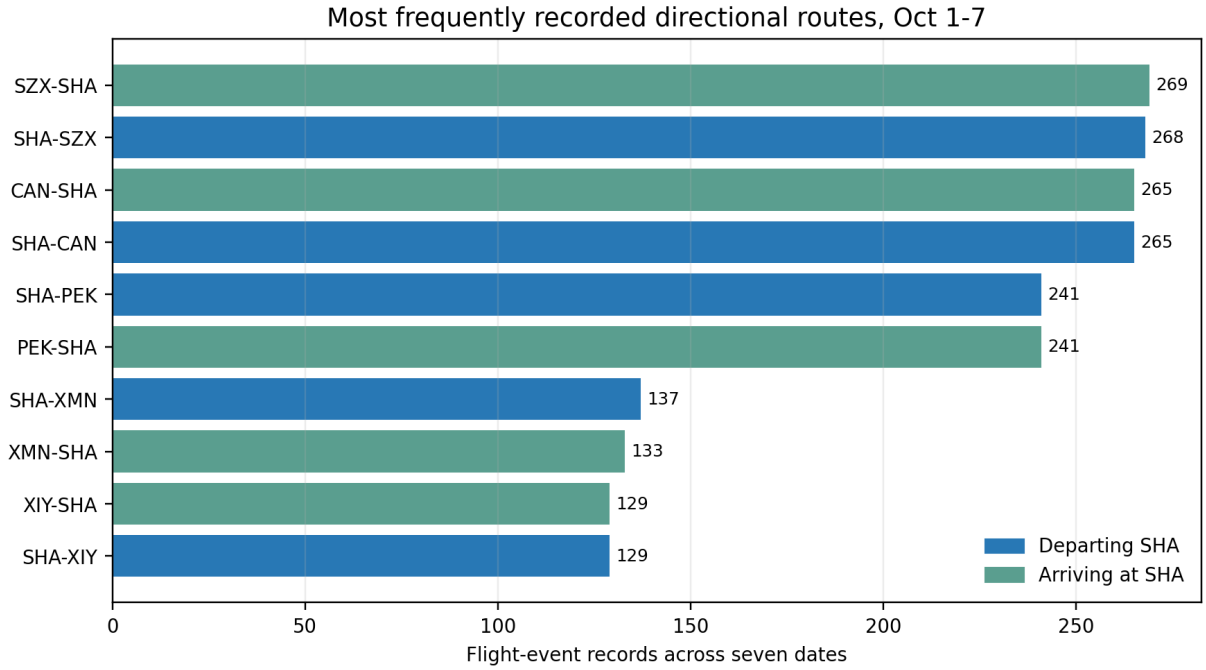


Figure 4. Ten most frequent directional routes in the October 1-7 records. SHA-origin routes are departures; routes ending in SHA are arrivals. A record count is not a passenger count.

The leading observed connections are Shenzhen-SHA and SHA-Shenzhen, followed by Guangzhou-SHA and SHA-Guangzhou, and the Beijing-SHA and SHA-Beijing pair. Directional labels are kept separate because an arrival and a departure on the same city pair are different service events. The chart describes event frequency in this airport file, not the number of unique travelers or the full size of each destination market.

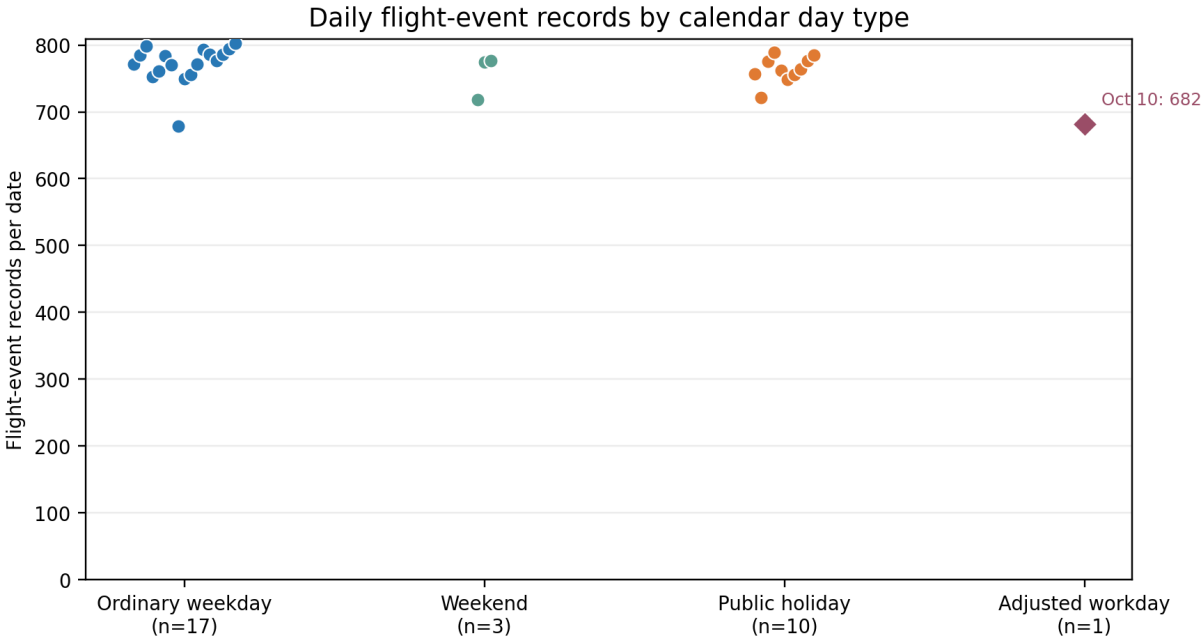


Figure 5. One point per observed local date, grouped by the dataset's calendar-day classification. n gives the number of dates in each category. The adjusted-workday category has one observation only (October 10).

The dataset marks Saturday, October 10 as an adjusted workday following the seven-day National Day holiday, consistent with the official 2026 holiday arrangement. That date has 682 flight-event records (342 arrivals and 340 departures); 30 of 680 known-status records are labeled canceled (4.41%). This observed count is below the sample's ordinary-weekend daily mean (756.7 across 3 dates), but only one adjusted workday is present, so the contrast cannot establish a general adjusted-workday effect. Importantly, the post-National Day seven-day comparison includes October 10; its mean and cancellation proportion therefore combine a make-up workday with ordinary weekdays and a Sunday.

The preceding comparison week also includes September 25-27, classified in the source calendar as the Mid-Autumn public holiday. Thus, the earlier period is not a fully ordinary baseline either. The three seven-day windows remain useful descriptive context, but differences between them may reflect the mix of official holidays, adjusted workdays, and ordinary days as well as other factors.

5. Discussion and planning implications

The data support a modest finding: recorded flight-event volume per day during October 1-7 is close to the immediately preceding weekday-aligned week, while the following week has fewer records per day. Cancellations are less frequent as a share of known statuses during the holiday week than in either neighboring window. These are observed associations in one short sample, not evidence that the holiday improved airport performance or that a particular policy caused the differences.

The Chinese make-up workday arrangement is a meaningful part of the travel-calendar context. October 10 falls on a Saturday but is designated an adjusted workday, rather than an ordinary weekend. Since the post-holiday comparison window includes this date and the dataset has no other adjusted workday, a

calendar-aware analysis should report it separately and avoid generalizing from it. Similarly, September 25-27 are Mid-Autumn public holidays, which limits the interpretation of the preceding period as a non-holiday baseline.

For airport operations and transportation planning, the results identify the value of calendar-aware monitoring: daily event counts, scheduled-hour shares, routes, cancellations, and the official work/rest calendar should be considered together. However, this file does not establish passenger volumes, terminal or runway capacity, or queues, and the reported patterns cannot be interpreted as realized airport throughput. Future operational analysis should connect scheduled services to verified movement times, capacity and disruption data.

A follow-up study could link verified flight movements to aircraft type and route distance, collect passenger volumes or defensible load-factor estimates, and compare available rail and air itineraries for selected corridors. Adding airport-rail access time, fares, and accessibility measures would connect the analysis to urban integration and social equity. Emissions could then be estimated with an explicitly documented method and uncertainty range. A longer multi-year sample and matched comparison dates would help separate holiday timing from weekday, season, weather, and disruption effects.

6. Limitations

The unit is a flight-event record, not a passenger or trip. Multiple records do not identify unique travelers, trip purpose, or mode choice.

The analysis covers one airport and 31 dates, with only one National Day period. Adjacent weeks are useful context, not a causal control group.

Only one adjusted workday (October 10) appears in the sample, so its flight-event count and status profile are a single-date case, not a representative estimate. The comparison windows also include Mid-Autumn and National Day public holidays.

Status labels may be incomplete or time-sensitive. The post-holiday comparison has more blank or unknown statuses, changing the observed cancellation denominator.

The dataset does not provide measured airport capacity, weather, air traffic control restrictions, aircraft rotations, passenger volumes, fares, mode alternatives, emissions, or traveler characteristics.

Timing semantics in the wider dataset are marked unverified. This report therefore avoids interpreting timestamp differences as validated delays.

The capture process and completeness of the source file cannot be independently verified from the supplied CSV alone.

7. Conclusions

This study documents flight-event patterns around China's October 1-7, 2026 National Day holiday at Shanghai Hongqiao Airport. In weekday-aligned seven-day comparisons, the holiday period averages 768.7 event records per day, compared with 764.4 in the preceding window and 732.3 in the following window. The observed known-status cancellation share is 1.91% during the holiday, versus 3.74% before and 2.41% after. The following window includes one adjusted workday, October 10, while the preceding window includes three Mid-Autumn public holidays; these calendar differences constrain period-to-period interpretation. Route, hourly, and calendar-type profiles add descriptive detail about service patterns. These results are useful for transportation planning, but are not estimates of passenger demand, causation, congestion, emissions, or traveler choice.

The principal contribution is a reproducible descriptive baseline paired with a calendar-aware interpretation of comparison windows. Broader assessment of sustainable aviation and surface-mode alternatives requires linked passenger, alternative-mode, equity, and emissions information. Future research should extend the observation period across multiple years, include comparable holiday and non-holiday dates, and link verified operational, passenger, and environmental measures.

References

Intergovernmental Panel on Climate Change. (2022). Climate Change 2022: Mitigation of Climate Change. Contribution of Working Group III to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change. Cambridge University Press. https://doi.org/10.1017/9781009157926

International Transport Forum. (2023). ITF Transport Outlook 2023. OECD Publishing, Paris. https://doi.org/10.1787/b6cc9ad5-en

General Office of the State Council of the People's Republic of China. (2025, November 4). Notice on the arrangement of some public holidays in 2026 (Guo Ban Fa Ming Dian [2025] No. 7). October 1-7 are the National Day holiday; September 25-27 are the Mid-Autumn Festival holiday; September 20 and October 10 are adjusted workdays. https://www.gov.cn/zhengce/content/202511/content_7047090.htm

User-supplied dataset. (2026). sha_master.csv [CSV file; flight-event records associated with Shanghai Hongqiao Airport].

Reproducibility: run `python scripts/generate_national_day_report.py` from the SHA-flight-data project directory. The script reads sha_master.csv and regenerates this PDF and its five figures.